

Backup, Restore, and Midterm Synthesis

Create an independent logical artifact, restore elsewhere, verify behavior, and integrate the PostgreSQL evidence cycle.

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A backup artifact is a claim about future recovery

The claim becomes evidence only after a separate restore works and the result is checked.

ARTIFACT

Known source, method, time, tool version, size, and checksum.

RESTORE

A different target receives the artifact without overwriting the source.

VERIFY

Structure, identity, rules, relationships, and behavior are tested.

RPO and RTO turn recovery language into objectives

Recovery point objective

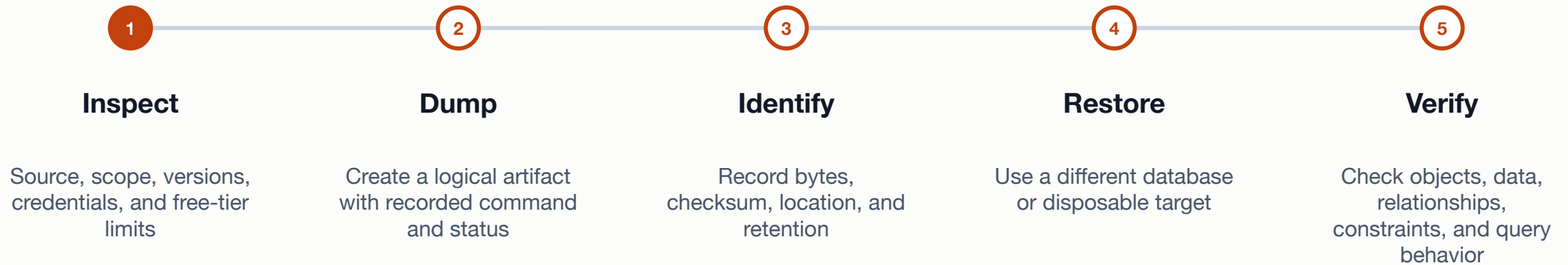
- Maximum tolerable data-loss window
- Drives artifact or log frequency
- Must match business and regulatory needs

Recovery time objective

- Target duration to restore acceptable service
- Includes detection, access, restore, verification, and cutover
- Must be tested with people and dependencies

Logical recovery is an ordered, verifiable workflow

Protect the source, identify the artifact, restore independently, and test behavior.



Tool compatibility is part of reproducibility

```
# Record server version
psql "$DATABASE_URL" -Atc "show server_version;"

# Create custom-format logical dump
pg_dump --format=custom \
  --file=metro_support.dump \
  "$DATABASE_URL"

# Identify exact artifact
shasum -a 256 metro_support.dump

# Restore to a different database
pg_restore --dbname="$RESTORE_URL" metro_support.dump
```

Do not publish URLs

- Use compatible client tools.
- Enter secrets through the environment or a secure prompt.
- Preserve redacted commands, status, size, and checksum.

Recovery verification needs independent kinds of evidence

- 1 Structure: expected schemas, tables, columns, views, constraints, and indexes exist.

- 2 Data: counts, stable identifiers, types, and selected values match expectations.

- 3 Relationships and rules: joins work and an invalid write still fails as intended.

- 4 Behavior and limits: a meaningful query succeeds; omissions and untested failures are named.

Lab: Create, restore, and verify a logical artifact

Use the recovery notebook to perform a real separate-target PostgreSQL restore without a cloud credential.

- Run the notebook from the first cell and record source, server/client versions, and dump command.
- Record artifact size and abbreviated checksum, then restore to the distinct target database.
- Complete structure, data, relationship, constraint, and meaningful-query checks.

SUBMIT ONE THING / one completed recovery notebook

The midterm integrates one PostgreSQL operating story

Build, observe, improve, recover, and communicate through one canonical case.

Four evidence threads make the midterm coherent

Build + query

Reproducible schema, constraints, seed data, and verified SQL question.

Meaning is explicit.

Operate

Transaction or blocking evidence identifies state, impact, mitigation, and verification.

Symptoms become diagnosis.

Improve + recover

One access/performance decision plus a logical recovery plan and checks.

Tradeoffs remain visible.

Cloud features and logical recovery cover different boundaries

Managed platform feature

- May automate snapshots, retention, or point-in-time mechanisms by plan
- Depends on current provider terms and configuration
- Still requires restore access, testing, and application recovery

Course logical artifact

- Portable schema and data representation under selected scope
- Can be created and tested without a paid backup feature
- Does not capture every provider or application setting

Audit the evidence before submitting

- 1 Every claim names its environment, identity, exact action, and observed result.

- 2 Stable identifiers, counts, expected failures, or plans make verification reproducible.

- 3 Only the canonical midterm requirements are used; no duplicate deliverable list is invented.

- 4 Credentials and private account details are absent; limitations and next checks are explicit.

Recovery completes the first half's evidence cycle

A dependable system protects meaning, controls change, reveals state, restricts access, measures work, and restores elsewhere.

- What exact artifact protects the data?

- Which separate restore proves usability?

- Which operating claim in the midterm has the strongest verification?